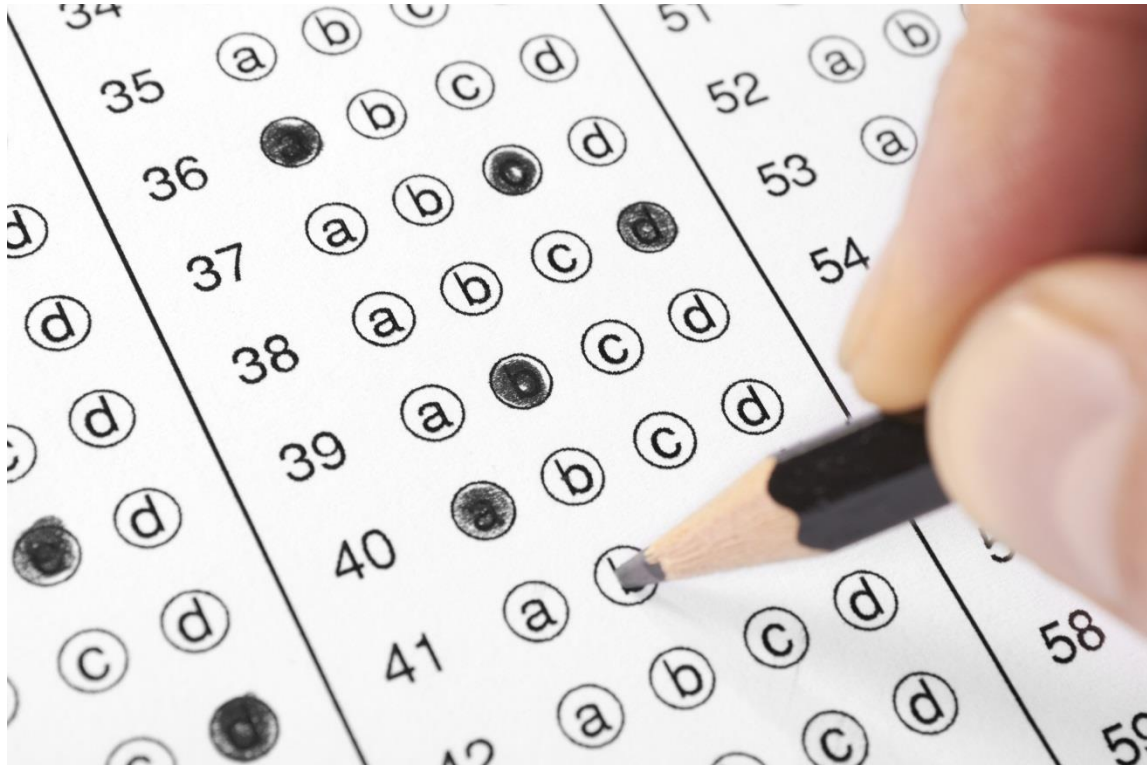


XI PHYSICS

MULTIPLE CHOICE QUESTIONS



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1.1: HISTORY OF PHYSICS

[2007, 2006, 2005, 2000, 2010]

1. KitabulManazir was written by:
a) Omar Khayyam b) Al-Beruni. c) Al-Razi d) **Ibn-al-Haitham.**

[2002]

2. The famous book Kitab-ul-QanoonMasoodi is written by:
a) Al-Masoodi b) **Al-Beruni.** c) Al-battni d) Al-Kindi

[2008]

3. Al Shifa, an encyclopedia of philosophy, was written by:
a) **Ibn-Sina** b) Omer Khayyam. c) YakoobKindi d) Al-Razi

4. Ibn-e-Sina was famous for his research in the field of:
a) Physics b) **Medicine** c) Mathematics d) None

[2002]

5. Ibn-e-Sina's book, which regarded a treatise on physics is:
a) **Al-Shifa** b) Algorith c) Ai-Manazir d) Al-Qanoon fit-tib

[2004]

6. Al-Qanoon fit-tib was written by:
a) Omer Khayyam b) Al-battni c) **Ibn-e-Sina** d) Al-Beruni

[2001]

7. The first book an optics was written by:
a) **Ibn-al-Haitham** b) Omar Khayyam. c) Al-Khawrizimi d) Al-Beruni

8. Laws of reflection and refraction are given by:
a) **Ibn-al-Haitham** b) Al-Beruni c) Al-Khawrizimi d) Omar Khayyam

[1996]

9. Pinhole camera was designed by:
a) Omar Khayyam b) Al-Beruni c) **Ibn-al-Haitham** d) Al-Khawrizimi

10. The density of metal was determined by:
a) **Al Beruni** b) Jabir-bin-Hayyan c) Dr. Abdul salam d) Yakoob Kindi

11. The founder of Analytical Algebra:
a) Al-Beruni b) Jabir-bin Hayyan c) Al-Bathani d) **Al-Khawarizimi**

12. The famous book Hisabuljubr-Wal-Muqabla is written by:
a) Al-Beruni b) Jabir-bin Hayyan c) Al-Battani d) **Al-Khawarizim**

[2009]

13. Screw and lever were invented by:
a) Newton b) Al Farabi c) **Archimedes** d) Galileo Galilei

[2014]

14. The Noble Prize in Physics was awarded to this Pakistani scientist:
a) Dr. Abdul Qadeer Khan b) Dr. SaleemuzZamanSiddiqui
c) **Dr. Abdus Salam** d) Dr. Samar Mubarak Mand

1.2: SYSTEM OF UNITS

1. The unit of electric current in S.I:
a) Watt b) Kwh c) Volt d) Ampere
[2014]
2. The unit of Luminous intensity is:
a) Decibel b) Candela c) Dioptred) Watt/m²
3. Candela is a unit of:
a) Mass b) Velocity c) Force d) Luminous intensity
[2003]
4. Thermodynamic unit of temperature is:
a) °F b) K c) °C d) All

1.3: SIGNIFICANT FIGURES

1. For the quantity 0.121203, the number of significant figure is;
a) 6 b) 7 c) 4 d) 3
[2003]
2. What is the number of significant figures in the number 50452:
a) 4 b) 2 c) 6 d) 5
3. What is the number of figures in the number 1.001110:
a) 6 b) 7 c) 4 d) 1
[2000]
4. What is the number of significant figures in the number 6.32
a) 3 b) 2 c) 4 d) 1
[2005]
5. The number 860.040 has:
a) 5 S.F b) 6 S.F c) 4 S.F d) 3 S.F
[2006]
6. The number of Significant figures in 2.050×10^6 is:
a) 2 b) 3 c) 4 d) 6
[2008]
7. The number of significant figures in 0.005040 is:
a) 3 b) 4 c) 5 d) 6
[2009]
8. The number of significant figure in 2.500×10^3 is:
a) 2 b) 3 c) 4 d) 7
[2010]
9. The number of significant figure of 7.050×10^{-2} is:
a) 2 b) 3 c) 4 d) 6
[2012]
10. The products of two numbers 5.642 and 4.71 in the prospective significant number is
a) 26.57382 b) 26.574 c) 26.6 d) 26.5738

1.4: DIMENSIONS

1. What are the dimensions of velocity?
a) LT^{-2} b) $L^{-2}T$ c) $L^{-1}T^{-1}$ d) LT^{-1}
2. What are the dimensions of acceleration?
a) LT^2 b) LT^3 c) LT^{-2} d) LT^1
- [2007]
3. What are the dimensions of momentum?
a) $ML^{-1}T$ b) $ML^{-2}T$ c) MLT^{-2} d) MLT^{-1}
4. What are the dimensions of force?
a) MLT^{-2} b) $ML^{-1}T^{-2}$ c) MLT^2 d) MLT^{-1}
- [2014, 2012, 2008]
5. The dimensions of torque are:
a) ML^2T b) ML^2T^2 c) ML^2T^{-2} d) MLT^2
- [2002]
6. What are the dimensions of angular velocity?
a) T^{-1} b) T^{-2} c) T^2 d) T
7. What are dimensions of angular acceleration?
a) T^{-1} b) T^{-2} c) T^2 d) T
- [2004]
8. What are the dimensions of work?
a) M^3LT^2 b) $ML^{-1}T$ c) ML^2T^{-2} d) MLT^{-1}
- [2012, 2002, 2009]
9. What are the dimensions of energy (kinetic energy)?
a) ML^2T^3 b) $ML^{-1}T$ c) ML^2T^{-2} d) MLT^{-1}
- [2006]
10. What are the dimensions of Power?
a) ML^2T^3 b) $ML^{-1}T$ c) ML^2T^{-3} d) MLT^{-1}
11. The dimension of density is:
a) LT^{-2} b) ML^3 c) $ML^{-1}T^{-2}$ d) ML^{-3}
12. What are dimensions of frequency?
a) T^2 b) T^{-3} c) T^{-1} d) T
- [2003, 2011]
13. The dimension of "G" are:
a) $M^{-1}L^{-2}T^3$ b) $M^{-1}L^3T^{-2}$ c) $ML^{-2}T^3$ d) none of these
- [2012, 2003, 2001, 2006, 2010]
14. The dimension of angular momentum:
a) ML^2T^{-1} b) ML^2T c) MLT^{-1} d) MLT^{-2}

2.1: SCALAR & VECTORS

1. A vector is a physical quantity which has:
a) Magnitude
c) Direction
b) Both magnitude and direction
d) Either magnitude or direction
 2. Scalar is a physical quantity:
a) Number only
c) Direction only
b) Number with proper units.
d) Number with direction
 3. A vector which can represent the position of a point with respect to some fixed point in a coordinate system is called.
a) null vector b) free vector **c) position vector** (d) None of these
- [2007]
4. A vector which can be displaced parallel to itself and is applied at any point is known as
a) Null b) Unit c) Position **d) Free**
- [2009, 2003, 1992]
5. If a vector quantity is divided by its magnitude, the vector obtained is called:
a) Unit vector b) position vector c) null vector d) free vector
- [1997, 1996, 1992]
6. If a Null or Zero vector is multiplied by a finite number, it gives a _____.
a) Unit vector **b) Zero vector** c) Negative vector d) None
 7. The magnitude of unit vector will always be:
a) 1 b) 0 (zero) c) -1 d) none of these
 8. The magnitude of a vector is always treated as:
a) negative **b) non-negative** c) Both a & b d) None
 9. Two or more vectors are added by:
a) head to tail rule b) simple addition c) Both a & b d) none of these
 10. The angle between the horizontal and vertical component of vector is:
a) 0° b) 45° **c) 90°** d) 180°
 11. The vector in space has:
a) One component **b) 3 components** c) Two components d) None of these
 12. The vector in a plane has:
a) One component **b) 2 components** c) Three components d) None of these
 13. When a vector is multiplied by a negative number its direction:
a) Remains same **b) changes by 180°** c) Not change d) Changes by 210°

2.2: ADDITION OF VECTORS

- [2009]
1. In addition of vectors by rectangular component method the direction ϕ of resultant vector is given by:
a) $\phi = \sin^{-1} (A_y/A_x)$ b) $\phi = \cos^{-1} (A_y/A_x)$ c) **$\phi = \tan^{-1} (A_y/A_x)$** d) None
 2. In parallelogram law of vector addition the resultant of the two vector is represented by:
a) diagonal of the parallelogram b) any adjacent side of the parallelogram
c) opposite side of the parallelogram d) None of these

3. Law of cosine is normally used to determine the:
a) **magnitude of resultant** b) direction of resultant
c) both magnitude & direction of resultant d) None of these
4. Law of Sine is normally used to determine the:
a) magnitude of resultant b) **direction of resultant**
c) both magnitude & direction of resultant d) None
5. A force of 5 N is acting along Y-axis. Its components along X-axis is:
a) 5N b) **Zero** c) 10N d) 5N
6. Two force act together on an object. The magnitude of their resultant is minimum when the angle between the force is.
a) 0 b) 45 c) 60 d) **180**
- [2008]
7. Two perpendicular vectors having magnitudes of 4 units and 3 units are added their resultant has the magnitude of:
a) 7 units b) 12 units c) 25 units d) **5 units**
- [2003]
8. If two forces of the same magnitude F makes an angle of 180° with each other. Their resultant is:
a) 2F b) **Zero** c) 0.5F d) F

2.3: DOT AND CROSS PRODUCTS

- [1996, 1997]
1. The dot product of two vectors is zero when they are
a) in the same direction b) **perpendicular to each other**
c) in the opposite direction d) None
2. The cross product of two vectors is zero when they are
a) parallel to each other b) perpendicular to each other
c) opposite in direction d) **both a) and c)**
3. The cross product of two vectors is a:
a) scalar b) **vector** c) Both a & b d) none of these
4. The dot product of two vectors is a:
a) **scalar** b) vector c) Both a & b d) none of these
5. The unit vectors $\hat{i}, \hat{j}$ and $\hat{k}$ are:
a) Parallel to each other b) **perpendicular to each other**
c) Opposite to each other d) none of the above
- [2006]
6. $\hat{k} \cdot (\hat{i} \times \hat{j})$ is equal to:
a) Zero b) **one** c) j d) $-k$
7. The dot product of a unit vector $\hat{i}$ & $\hat{k}$ is:
a) **Zero** b) 1 c) -1 d) j
- [2001]
8. $\hat{k} \cdot (\hat{i} \cdot \hat{j})$ has value:
a) **Zero** b) One c) j d) k
- [2002]
9. $\hat{j} \times \hat{j}$ is equal to:

- a) j^2 b) j c) one **d) zero**
10. If i, j and k are the unit vectors along x, y and z axes respectively, then $k \times j =$
a) $\hat{i}$ b) $-\hat{i}$ c) 1 d) -1

[2012]

11. If $\vec{A} = a\hat{i}$ and $\vec{B} = b\hat{j}$, then $\vec{A} \times \vec{B}$ is equal to:
a) $\vec{0}$ b) $a b \hat{k}$ c) $-ab\hat{k}$ d) none of these

[2014 ,2005]

12. If $\vec{A} \cdot \vec{B} = 0$ and $\vec{A} \times \vec{B} = 0$ and $\vec{A} \neq \vec{0}$, the vector $\vec{B}$ is:
a) Equal to A **b) Zero** c) Perpendicular to A d) Parallel to A

[2004]

13. If $\vec{A} \cdot \vec{B} = 0$ when, $\vec{A} \neq \vec{0}$, $\vec{B} \neq \vec{0}$, then the vectors are:
a) Parallel b) Opposite **c) Perpendicular** d) None

[1996, 1997 , 2003]

14. If $|\vec{A} \times \vec{B}| = \vec{A} \cdot \vec{B}$ then angle between vector $\vec{A}$ and $\vec{B}$ is:
a) 0° **b) 45°** c) 90° d) 180°
15. $|\vec{A} \times \vec{B}| = 0$ and $|\vec{A} \cdot \vec{B}| = 0$ then the vectors $\vec{A}$ and $\vec{B}$ are:
a) parallel to each other **b) either A or B is a null vector**
c) perpendicular to each other d) None
16. If $\vec{A}$ and $\vec{B}$ are two vectors, which of the following is correct.
a) $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$ b) $\vec{A} \times \vec{B} = \vec{B} \times \vec{A}$ c) $\vec{A} \cdot \vec{B} = -\vec{B} \cdot \vec{A}$ d) none of these

[2007]

17. The area of a parallelogram formed of two vectors A and B is given by:
a) $\frac{1}{2} (\vec{A} \cdot \vec{B})$ **b) $|\vec{A} \times \vec{B}|$** c) $\frac{1}{2} |\vec{A} \times \vec{B}|$ d) $(\vec{A} \cdot \vec{B})$

[2004]

18. When $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, the angle between the vectors A and B is
a) Zero b) 45° **c) 90°** d) 180°
19. If $\vec{F} = 6\hat{i} - \hat{j}$ and $\vec{d} = 2\hat{i} + 2\hat{j}$ the work done will be:
a) 0 b) 6 J **c) 10 J** d) 12 J

[2014]

41. If $\vec{A} = 5\hat{i} + \hat{j}$ and $\vec{B} = 2\hat{k}$ then $\vec{A} - \vec{B}$ is equal to:
a) $5\hat{i} + \hat{j} + 2\hat{k}$ b) $5\hat{i} - \hat{j} - 2\hat{k}$ c) **$5\hat{i} + \hat{j} - 2\hat{k}$** d) $-5\hat{i} - \hat{j} + 2\hat{k}$

1. Two objects moving with same velocities are _____ with respect to each other:
a) **At rest** b) In motion
c) In uniform motion d) None of these
2. Rate of change of displacement is called
a) Speed b) **Velocity**
c) Acceleration d) None of these

[2013]

3. When value of average and instantaneous velocities are equal, body is said to be moving with:
a) Uniform speed b) **Uniform velocity**
c) Uniform acceleration d) Average velocity
4. When the values of average and instantaneous acceleration are equal, the body is said to be moving with:
a) Average acceleration b) **Uniform acceleration**
c) Positive acceleration d) None of these
5. The acceleration of a body moving with uniform velocity is:
a) **Zero** b) No
c) Not uniform d) None of these
6. Rate of change of velocity is called
a) **Acceleration** b) uniform acceleration
c) Variable velocity d) None of these
7. Body A is moving with velocity 30 m/s towards east and body B is moving with velocity 20 m/s towards west, the relative velocity of A with respect to B is.
a) 10 m/s b) **50 m/s**
c) 30 m/s d) 20 m/s

[1989]

8. Two stones A and B are thrown from the top of a tower. The stone "A" is thrown vertically upward while the stone "B" is thrown vertically downward with the same speed. Which of the following statements is true:
a) Stone A strikes the ground with a higher velocity than that of "b"
b) Stone B strikes the ground with a higher velocity than that of "A"
c) **Both strike with the same velocity** d) None of these
9. A ball 'A' dropped from the top of a building while another ball 'B' is thrown horizontally at the same time. Which ball strikes the ground first?
a) Ball 'A'
b) Ball 'B'
c) **Both strikes simultaneously** d) None of these
10. If a weight of 1kg and 1gm are dropped from the same height simultaneously then:
a) 1kg will reach the ground earlier b) 1gm will reach the ground earlier
c) **Both will reach at the same time** d) None of these

[2002]

11. If the velocity time graph of a body is a straight line, parallel to time- axis, its acceleration is
a) **Zero** b) Maximum
c) Minimum d) Uniform

[2005]

12. If the velocity time graph of a moving body is a curve, the body moves with:
- a) Constant speed
 - b) Constant velocity
 - c) Constant acceleration
 - d) **Changing acceleration.**

[2007]

13. When a constant force is applied on a body, it move with:
- a) Constant speed
 - b) Constant velocity
 - c) **Constant acceleration**
 - d) None
14. Swimming is possible because of which law of motion
- a) First
 - b) Second
 - c) **Third**
 - d) None of these
15. A boy pushes first a large and then a small car. He moves the small one faster. This is an example of:
- a) Newton's first law of motion
 - b) **Newton's second law of motion**
 - c) Newton's third law of motion
 - d) None of these
16. Newton's first law of motion is also called
- a) Law of torque
 - b) Law of force
 - c) **Law of inertia.**
 - d) None of these
17. The second law of motion shows the relation between
- a) Mass and weight
 - b) Mass and velocity
 - c) **Mass and acceleration**
 - d) Force and Velocity
18. Newton's second law of motion is also called:
- a) Law of gravitation
 - b) Law of inertia
 - c) Law of inertial frame
 - d) **Law of acceleration**
19. One dyne is equal to:
- a) **10^{-5} N**
 - b) 10^5 N
 - c) 10^{-3} N
 - d) 10^3 N

3.2: MOMENTUM

[2006 , 04]

20. The rate of change of linear momentum is equal to:
- a) Acceleration
 - b) Torque
 - c) Angular velocity
 - d) **Force**
21. The unit of linear impulse is the same as that of:
- a) Force
 - b) **Momentum**
 - c) Energy
 - d) None of these
22. Impulse or Impulse of force can be expressed as:
- a) $F \Delta t$
 - b) $m v_f - m v_i$
 - c) Change of momentum
 - d) **All of these**

[2012]

23. A helicopter weighing 3920 N if is moving up with the constant speed of 4m/s. the force on helicopter is
- a) 4720N
 - b) **3920N**
 - c) 3924N
 - d) 3916N

[2002]

24. A helicopter of mass 3×10^3 kg rises vertically with a constant speed of 25 m/sec. what resultant force acts on the helicopter?
- a) Zero
b) 3×10^4 N downward
c) **3×10^4 N upward**
d) 7.5×10^4 N upward.

[2003]

25. A body goes from 2 meter to 8 meter mark and back to 2 meter mark in 3 sec. its average speed is:
- a) 2 m sec^{-1}
b) 6 m sec^{-1}
c) **4 m sec^{-1}**
d) Zero

[2003 (P.M)]

26. How much height does a freely falling body of mass 10 kg lose in 2 sec?
- a) 9.8 m
b) **19.6 m**
c) 49 m
d) 4.9 m

[2002 P.M , 2011]

27. A 1kg stone when falling from a height of 10 m, strikes the ground with the velocity of
- a) 10 m/s
b) **14 m/s**
c) 98 m/s
d) 19.6 m/s
28. A body has an initial velocity of 8 m/s. after moving 4m its velocity is 12 m/s. its acceleration?
- a) **10 m/s^2**
b) 100 m/s^2
c) 4 m/s^2
d) 40 m/s^2

[2010]

29. How many meter(s) will a 20 Kg ball, starting from rest, fall freely in one second?
- a) **4.9 m**
b) 9.8 m
c) 4.0 m
d) 19.6 m

[2013]

30. A body fall freely the distance covered by it in 2 sec is?
- a) **19.6 m**
b) 9.8 m
c) 39.2 m
d) 100 m
31. The quantity of matter in a body is called
- a) Momentum
b) Force
c) **Mass**
d) Velocity
32. The quantity of motion in a body is called its
- a) **Momentum**
b) Force
c) Mass
d) Velocity

[2003 (P.E)]

33. The unit of liner momentum is
- a) N / S
b) **NS**
c) JS
d) J / S

[2008 , 01]

34. In an inelastic collision of two bodies
- a) K.E. is conserved only
b) **Momentum is conserved only**
c) Both K.E and momentum are conserved
d) Neither K.E nor momentum is conserved

[1990]

35. A bullet is fired from a rifle. If rifle recoils freely, state whether the kinetic energy of rifle is:
- a) Greater than
b) Equal to
c) **Less than that of the bullet**
d) None of these
36. When a bullet is fired the gun move back
- a) To maintain the force
b) To enhance the velocity of bullet
c) **To keep the momentum conserves**
d) None of these

3.3: FRICTION

37. The kinetic friction will always be:

- a) greater than the static friction
- b) Equal to the static friction
- c) Less than the static friction
- d) None of these

[2012]

38. The property of fluids due to which they resist their flow is called:

- a) Static friction
- b) Coefficient of friction
- c) Viscosity
- d) Terminal velocity

[2007, 06, 02 P.E]

39. Stokes' Law for fluid friction is given as:

- a) $F = 6 \pi \eta r^2 v$
- b) $F = 6 \pi \eta r v$
- c) $F = 4 \pi \eta r^2 v$
- d) $F = \pi \eta r v$

[2005]

40. In Stoke's Law, the viscous force is not proportional to:

- a) Co-efficient of viscosity
- b) Radius of the sphere
- c) Terminal velocity
- d) Mass of the sphere.

[2002 (P.M), 2010]

41. Stoke's Law holds goods for the:

- a) Bodies of all shapes
- b) Motion through viscous medium
- c) Motion through non-viscous medium
- d) Motion through a vacuum

[1996]

42. A rain drop continues to fall with a uniform velocity when:

- a) Its weight is balanced by air friction
- b) weight is balanced by air friction & up thrust
- c) Its weight is balanced by up thrust
- d) None of these

5.1: TORQUE, ANGULAR MOMENTUM & EQUILIBRIUM

[2008]

1. The centre of the mass of a system of particles:
- a) Always coincides with the centre of gravity
 - b) Never coincides with the centre of gravity.
 - c) Coincides with the centre of gravity in a uniform gravitational field.**
 - d) Coincides with the centre of gravity in a non- uniform gravitational field.

[2008]

2. Which of the following is a spin motion?
- a) The motion of the planets round the sun
 - b) The motion of electrons round the nucleus
 - c) The motion of the moon round the earth
 - d) Daily rotation of the earth causing day & night.**

[2007 , 2005 , 2002 (P.E)]

3. The rate of change of angular momentum with respect to time is:
- a) Force
 - b) Angular velocity
 - c) Angular acceleration
 - d) Torque**

[2006]

4. Two forces which are equal in magnitude but opposite in direction, not acting on the same line, constitute a:
- a) Couple**
 - b) Circle
 - c) Power
 - d) Force

[2006]

5. The S.I unit of angular momentum is:
- a) kg m/s
 - b) kg ms
 - c) J-s**
 - d) J-s⁻¹

[2005]

6. A body in equilibrium
- a) Can move with constant acceleration
 - b) is always at rest
 - c) Can move with constant velocity**
 - d) None of these

[2004]

7. When the net torque acting on a system is zero, which of the following will be constant?
- a) Force
 - b) Angular momentum**
 - c) Linear momentum
 - d) All of these

[2003 (P.E)]

8. The physical quantity which produce angular acceleration is called:
- a) Torque**
 - b) Work
 - c) Power
 - d) Energy

[2002 (P.E)]

9. A body may be in equilibrium when:
- a) It is in motion**
 - b) It is moving with a uniform acceleration.
 - c) It is the rest
 - d) It is moving with a variable velocity.

[2002 (P.E)]

10. The absolute P.E. of a body of mass “m” in the earth’s gravitational field is given by:
- a) - G M_e m / r**
 - b) - G M_e m / r²
 - c) G M_e m / r
 - d) G M_e m / Re²

[2001]

11. Every point of a rotating rigid body has the same:

- a) Linear acceleration
- b) Angular velocity
- b) Linear momentum
- d) Linear velocity

[1992]

12. Torque is the vector product of displacement vector and _____ vector.

- a) momentum
- b) Force
- c) torque
- d) All of these

13. The magnitude of torque of the couple depends on:

- a) the distance of $(\vec{F})$ from origin
- b) the distance of $(-\vec{F})$ from origin
- c) the distance between $(\vec{F})$ and $(-\vec{F})$
- d) none of these

14. If the axis of rotation of a rotating body passes through the body itself, then its motion is called:

- a) Linear motion
- b) Orbital motion
- c) Spin motion
- d) S.H.M motion

[2008]

11. If the mass of the earth becomes four times to its initial value then the value of 'g' will be:
a) Equal to its initial value
b) **Four times to its initial value**
c) One fourth of its initial value
d) All of these
12. The mass of a planet and its diameter are three times that of earth then the acceleration due to gravity on the surface of the planet will be:
a) **One third on the earth's**
b) Half on the earth's
c) One forth on earth
d) None of the these
13. If a planet existed whose mass and radius were both twice that of the earth, then acceleration due to gravity at its surface would be:
a) **4.9 m/s²**
b) 19.6 m/s²
c) 2.45 m/s²
d) All of these

[1990]

14. If the radius of earth were to shrink its mass remaining the same, the acceleration due to gravity on the earth's surface would
a) Decrease
b) **Increase**
c) Remain the same.
d) All of these

[2003 (P.M) 90]

15. If the radius of the earth were to shrink by 1% while its mass remaining same, the acceleration due to gravity on the earth's surface would:
a) Decrease
b) Remain the same
c) **Increase**
d) All of these
16. What happens to the person's weight when he goes down to the bottom of deep mine compared to his weight on the surface?
a) It will remain the same
b) It will increase
c) **It will decrease**
d) All of these
17. The acceleration of free fall on moon is about one sixth of its value on earth. If on the earth a body has mass 'm' and weight 'w', then on the moon, mass and weight will be respectively about:
a) $\frac{m}{6}$ and $\frac{w}{6}$
b) $\frac{m}{6}$ and w
c) **m and $\frac{w}{6}$**
d) All of these
18. On the surface of the moon the weight of a person:
a) Increases
b) **Decreases**
c) Remains the same
d) All of these
19. A person whose weight is 120 pound on the earth, on the moon his weight will be approximately:
a) **20 pound**
b) 30 pound
c) 40 pound
d) All of these

[2013]

20. The weight of a man is 600N at the earth; his weight on the moon, where $g_m = \frac{g}{6}$, will be:
a) 3600N
b) 600N
c) 300N
d) **100N**

21. If the distance between two masses is doubled, the gravitational force between them becomes:
a) **One fourth of its original value** b) Half of its original value
c) Four times of its original value d) All of these
- [2014]
22. Artificial gravity can be created in the spaceship by producing:
a) translatory motion b) vibratory motion
c) **spin motion** d) orbital motion
- [2006 , 01]
23. The weight of an object in a satellite orbiting around the earth is:
a) Actual weight b) Greater than the actual weight
c) Less than the actual weight d) **Zero**
24. A spring balance is being used to weigh a mass of 1 kg in a lift. If the spring balance reads 9N and the acceleration of free fall $g = 10 \text{ m/s}^2$, the lift is :
a) Ascending at 1 m/s^2 b) At rest
c) **Descending at 1 m/s^2** d) All of these
- [2002 (P.M)]
25. Artificial gravity is produced in a space – craft to:
a) Increases the weight of the astronaut b) Decrease the weight of the astronaut
c) **Overcome the state of weightlessness.** d) Overcome the force of friction
26. Spring balance is used to measure
a) **Apparent weight of the object** b) Mass of the object
c) Torque of body d) None of the these
- [2001]
27. The apparent weight of a person standing in an elevator which is moving down with uniform acceleration will be:
a) Same as that on the surface of the earth
b) Greater than its weight on the surface of the earth
c) **Less than its weight on the surface of the earth**
d) All of these

7.1: WORK POWER & ENERGY

1. The dot product of force and displacement is:
a) **Work**
c) Power
b) Energy
d) All of these
 2. When the force and displacement are parallel to each other then work is:
a) Minimum
c) Opposite
b) **Maximum**
d) All of these
 3. When the force and displacement are perpendicular to each other then work is:
a) **Minimum**
c) Opposite
b) Maximum
d) All of these
 4. When the force and displacement are in the opposite direction then work is:
a) Positive
c) Zero
b) **Negative**
d) All of these
 5. The rate of change of momentum multiplies by displacement gives:
a) Power
c) **Work**
b) Pressure
d) All of these
 6. A person having a mass of 60 kg exerts a horizontal force of 300 N in pushing a 90 kg object through a distance of 3m along a horizontal floor. The work done by this person is:
a) 7000 Joules
c) 1100 Joules
b) **900 Joules**
d) All of these
 7. Work done in the gravitational field:
a) **Is independent of the path followed**
c) Both a & b
b) Depends upon the path followed
d) None of these
- [2007]
8. The work done by a conservative force along a closed path is:
a) Positive
c) **Zero**
b) Negative
d) None
- [2008]
9. The work done by the centripetal force "F" on a body moving in a circle is zero because:
a) The body moves parallel to F.
c) **The body moves right angle to F**
d) The centripetal and centrifugal forces balance each other.
b) The body moves opposite to F.
 10. A man do the work if he:
a) **Goes the fifth floor of the building**
b) Remains on the ground floor
c) Goes to fifth floor of the building and comes back to ground floor.
d) All of these
- [2007]
11. The dot product of force and velocity is called?
a) Work
c) Energy
b) **Power**
d) All of these
- [2005]
12. Power is equal to:
a) $F \times d / t$
c) **$F \cdot d / t$**
b) $F \cdot V / t$
d) $F \times V / t$
 13. A man weighing 600 N climbs 5.0 m vertically upward in 8.0 seconds his rate of working
a) 175 watt
c) **375 watt**
b) 275 watt
d) All of these

14. Watt may be defined as:
a) Joule per coulomb
c) Newton meter
b) **Joule per second**
d) All of these
- [2003 (P.E)]
15. One horse power is equal to:
a) **550 ft. lb/s**
c) 720 ft/Lb
b) 746 ft. lb/s
d) None of the these
[2014, 05]
16. If mass and speed both are doubled, the kinetic energy of a moving body.
a) Increases 4 times
c) **Increases 8 times**
b) Increase 6 times
d) Remains the same.
[2014, 04, 2003 (P.E)]
17. Kilo watt hour is a unit of
a) **Energy**
c) Time
b) Power
d) Force
- [2001]
18. Which of the following is not the unit of power?
a) **Kilowatt hour**
c) footlbs/sec
b) Horse power
d) Joules/sec
19. One eV is equal to?
a) **$1.6 \times 10^{-19} \text{ J}$**
c) $1.6 \times 10^{19} \text{ J}$
b) $1.6 \times 10^{-18} \text{ J}$
d) All of these
20. Law of conservation of energy is equivalent to:
a) **Law of conservation of mass**
c) Law of Inertia
b) Law of conservation of Momentum
d) All of these
21. Work energy equation is simply:
a) Law of conservation of mass
c) Law of inertia
b) **Law of conservation of Energy**
d) None of these
22. A ball during its downward journey possesses.
a) Kinetic energy only
c) **Both K.E & P.E**
b) Potential energy only
d) All of these
23. Einstein's mass energy equation is:
a) $E = m c$
c) $E = m^2 \cdot c$
b) **$E = m c^2$**
d) All of these
24. The tidal energy is due to the :
a) **Rotation of earth relative to moon**
c) Both a & b
b) Rotation of the earth around sun
d) None of these
25. Which of the following represents the energy lost by a 1.0 N weight in falling through 1.0m:
a) 0.10 J
c) 10.0 J
b) **1.0 J**
d) None
26. Which of the following has the same unit as that of potential energy:
a) Acceleration
c) **Work**
b) Momentum
d) All of these
27. What is the kinetic energy of 60 g bullet moving at a speed of 600 m/s:
a) **10, 800 J**
b) 11,800 J
c) 12,800 J
28. Which of the following does not convert one type of energy into another:
a) Solar cell
b) Steam engine
c) **Transformer**
- [2013]
29. A weight lifter consumes 500J of energy to lift a load in 2 seconds, the power used by him
a) 125 watt
b) **250 watt**
c) 500 watt
d) 1000 watt

8.1: WAVES, SOUND & MOTION

1. [2009]
A body executes simple harmonic motion if:
(a) $a = k \cdot x$ (b) $v = -k \cdot x$ (c) $a = -\sqrt{k} \cdot x$ (d) $a = -k \cdot x^2$
2. [2004]
The value of elastic restoring force in case of a spring is:
(a) Kx (b) $-Kx$ (c) $\frac{1}{2}Kx$ (d) None of these
3. [2008]
The frequency of a simple pendulum is given by
(a) $\nu = 2\pi\sqrt{g/L}$ (b) $\nu = 2\pi\sqrt{L/g}$ (c) $\nu = 1/2\pi\sqrt{L/g}$ (d) $\nu = 1/2\pi\sqrt{g/L}$
4. [2008]
If the mass of a body in a spring is increased to 4 times, the period of vibration will be:
(a) 4times (b) 2 times (c) $\sqrt{2}$ times (d) Same as before
5. [2007 , 02 P.M]
If the mass of a body in a spring is doubled, the period of vibration of the body becomes:
(a) Double (b) Half (c) $\sqrt{2}$ times (d) four times
6. [2005]
If the bob of a vibration simple pendulum is suddenly detached from the string at its mean position, its path will be:
(a) A straight line (b) A circle (c) A parabola (d) A hyperbola
7. [2013, 2003 P.M]
The time period of a simple pendulum depends upon its
(a) Length (b) Amplitude (c) Mass of the bob (d) Temperature
8. [2003 P.M]
An object is executing simple harmonic motion. Its kinetic energy is maximum at its:
(a) Mean position (b) Extreme position (c) Extreme position. (d) At any point
9. [2004]
The frequency of a second's pendulum is:
(a) 1 Hz (b) 2 Hz (c) 0.5 Hz (d) None of these
10. [2004 , 2003 P.E]
If the bob of a simple pendulum is replaced by another bob of double mass but of the same size, its time period.
(a) Increases (b) Decreases (c) Remains the same (d) Becomes infinity
11. [2002 (P.E.)]
The frequency of oscillation of a simple pendulum depends upon:
(a) Mass of the bob (b) Amplitude (c) Length (d) None of them
12. [2002 (P.E.)]
Acceleration of free fall on the moon is $1/6$ of that on the earth. What would be period on moon of simple pendulum which has a period of 1 second on the earth?
(a) $1/6$ second (b) $1\sqrt{6}$ second (c) $\sqrt{6}$ seconds (d) 6 seconds
13. [2010]
Which of the following does not exhibit Simple Harmonic Motion?
(a) A hanging spring supporting a weight (b) The balance wheel of a watch
(c) The wheel of an automobile (d) The string of a violin

14. If a body oscillates with 0.7 KHz, then its angular velocity would be:
 (a) 4396 rad/sec (b) 4458 rad/sec (c) 2580 rad/sec (d) 4000 rad/sec
15. **[2012]**
 $\sin \theta = \theta$ if θ is specifically less than:
 (a) 15° (b) 10° (c) 5° (d) 1 radian
16. **[2005]**
 The velocity of a sound in a gas increases with:
 (a) Temperature (b) Pressure (c) Loudness (d) Frequency
17. **[2003 P.M]**
 Which one of the following properties of sound is affected by the change in temperature?
 (a) Amplitude (b) Wavelength (c) Frequency (d) Intensity
18. **[2008 2002, (P.E.)]**
 When the temperature of air rises, the speed of sound waves increases because.
 (a) The frequency of the wave increases. (b) The wavelength of the wave increases.
 (c) Both the frequency and wavelength (d) Neither frequency nor wavelength increase
19. **[2006]**
 Which of the following represents longitudinal waves?
 (a) Light waves (b) Sound waves (c) Radio waves (d) X-rays
20. **[2007, 2009]**
 The speed of sound is the highest in:
 (a) Solid (b) Liquid (c) Gas (d) Vacuum
21. **[2004]**
 Which of the following are compressional waves
 (a) Light waves (b) Sound waves (c) Radio waves (d) None of these
22. Which of the following is not the property of sound waves?
 (a) Interference (b) Diffraction (c) Polarization (d) Refraction
23. Speed of sound waves does not depend upon:
 (a) Wind speed (b) Temperature (c) Pressure (d) Density of the medium
24. **[2010]**
 The velocity of sound in space is:
 (a) 332 m/sec (b) 344 m/sec (c) 330 m/sec (d) Zero
25. The velocity of sound at 0°C is 332m/s. the increase in velocity for one degree rise of temperature in air is:
 (a) 61 m/s (b) 68 cm/s (c) 61 cm/sec (d) 60 cm/s
- [2014, 11]**
26. The earth quake waves are the example of:
 (a) Audible waves (b) Infrasonic waves (c) Shock Waves (d) Ultrasound waves
27. **[2003 P.E.]**
 Supersonic waves have the frequency of more than:
 (a) 20 Hz (b) 2000 Hz (c) 5000 Hz (d) 20000 Hz
28. **[2005, 02 P.M]**
 Human beings can hear sound waves having frequency:
 (a) 5 Hz (b) 5000Hz (c) 25000 Hz (d) 50000Hz
29. **[2003 P.M.]**
 The physical quantity which is related to loudness of sound is:
 (a) Frequency (b) Intensity (c) Quality (d) Wavelength
30. One Sone at 1000 H z is equal to:
 (a) 60 dB (b) 40 dB (c) 30 dB (d) 100 dB

31. **[2003 P.E.]**
The intensity level of a sound of intensity 10^{-12} watt/m² in a bell is
(a) Zero (b) One (c) Two (d) Three
32. **[2002 (P.M.)]**
The relative intensity (**I/I₀**) of the sound of a jet engine is 10^{13} . The intensity level will be.
(a) 13 (b) 130 (c) 1300 (d) 1.3
33. Intensity level of ordinary conversation (10^{-6} watt/m²) is:-
(a) 40 db (b) 60 db (c) 50 db (d) 70 db
34. The S.I. unit of intensity of sound is:
(a) Watt / m² (b) decibel (c) weber (d) dioptré
35. **[2012, 07, 2003 P.E.]**
The S.I unit of intensity level of sound is:
(a) Watt (b) Diopter (c) sone (d) Decibel
36. **[2007, 2010]**
The pitch of the sound depends on:
(a) Frequency (b) Wavelength (c) Intensity (d) Velocity
37. **[2007, 2002 (P.E.), 99]**
We recognize the voice of friends over the telephone by the _____ of sound.
(a) Quality (b) Intensity (c) Loudness (d) Pitch
38. **[2003 P.E.]**
When two exactly similar waves travel in a medium in opposite direction, they produce:
(a) Standing waves (b) Beats (c) Resonance (d) Diffraction
39. **[2001]**
The distance between node and anti node in a loop is
(a) λ (b) $\lambda/2$ (c) $\lambda/3$ (d) $\lambda/4$
40. **[2014, 2006]**
The distance between two consecutive nodes of stationary waves is:
(a) λ (b) $\lambda/2$ (c) $\lambda/4$ (d) $\lambda/6$
41. Frequencies which are multiples of fundamental frequency are called:
(a) Harmonics (b) Doppler frequencies (c) Beat frequencies (d) Nodal frequencies
42. **[2003 P.M, 2009]**
If the tension in a string is made four times, the speed of transverse waves will be:
(a) Half (b) Doubled (c) Four time (d) Same
43. **[2002, (P.M.)]**
If a string is vibrating in its fundamental mode, then to excite to the next higher mode, the tension must be:
(a) Halved (b) Doubled (c) Tripled (d) Quadrupled
44. **[2008]**
If you go on increasing the stretching force on a wire in guitar, its frequency of vibration
(a) Increases (b) Decrease (c) Becomes zero (d). Remain same
45. **[2008]**
If fundamental frequency of a string fixed at both ends is 50Hz, the fourth harmonic will be:
(a) 100 Hz (b) 150Hz (c) 200 Hz (d) 250 Hz
46. **[2002,(P.E.)]**
When stationary waves are set up in a stretched string it has fundamental frequency of 1,000 Hz. What would be the new fundamental frequency if tension is increased 4 times?
(a) 980 Hz. (b) 500 Hz. (c) 1010 Hz. (d) 2000 Hz.

47. For good acoustics of a hall, reverberation should not be:
(a) Too small. (b) Too large
48. **[2002 (P.M.)]**
As a source of sound moves away from a stationary listener, there is an apparent:
(a) Increase in pitch (b) Decrease in wavelength (c) Decrease in phase (d) Decrease in pitch
49. **[2000]**
When speed of jet planes becomes greater than the speed of sound then the loud sound produced is due to.
(a) Shock wave (b) Standing wave.
50. **[2002 (P.M.)]**
Supersonic jets may produce a double sonic boom due to two shock waves within the time interval of:
(a) 0.02 sec (b) 0.1 sec (c) 0.001 sec (d) 0.21 sec
51. **[2002 (P.E.)]**
A 252 Hz tuning fork produces four beats per second when sounded with another tuning fork of unknown frequency. What are the two possible values for the unknown frequency?
(a) 63 Hz. (b) 256 Hz and 248 Hz (c) 1008 Hz. And 1004 Hz (d) 248 Hz
52. **[2011, 2004]**
If two tuning forks of 256 Hz and 260 Hz are sounded together, the number of beats per second will be
(a) 3 (b) 4 (c) 5 (d) None of these
53. **[2013, 2007]**
The maximum number of beats that can be detected by human ear is:
(a) 3 (b) 5 (c) 7 (d) 9
54. **[2008, 06 , 03 P.E]**
Beats are produced due to:
(a) Diffraction (b) Interference (c) Polarization (d) Refraction
55. **[2003 P.M]**
When two vibrating bodies have slight different frequencies, they produce:
(a) Echo (b) Beats (c) Resonance (d) Polarization
56. **[2012]**
The Frequency of wave produced in a stretched string depends upon:
(a) length (b) tension (c) linear density (d) all of these

ANSWER KEY

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
c	B	d	b	c	C	a	A	c	c	c	c	c	a	C
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	b	b	b	a	b	c	c	d	c	b	d	b	b	b
31	32	33	34	35	36	37	38	39	40	41	42	43	44	45
a	b	b	a	d	a	a	a	D	B	a	b	d	a	C
46	47	48	49	50	51	52	53	54	55	56				
d	a	d	a	a	b	b	c	b	b	d				

9.1: NATURE OF LIGHT

1. Light wave:
(a) Require air or another gas to travel through
(b) Require an electric field to travel through
(c) Require a magnetic field to travel through
(d) **Can travel through a perfect vacuum**
2. Who proposed electromagnetic wave theory?
(a) Fresnel (b) Huygen (c) **Maxwell** (d) Fraunhofer
[2012]
3. According to Maxwell's theory, light travels in the form of _____ wave?
(a) transverse (b) Longitudinal (c) Mechanical (d) **Electromagnetic**
4. Maxwell based his theory of electromagnetic waves on the hypothesis that a changing electric field gives rise to:
(a) An electric current (b) Stream of electrons
(c) **A magnetic field** (d) longitudinal waves
[2011, 04]
5. Electromagnetic waves consist of oscillatory electric and magnetic field. Both fields are:
(a) Parallel to each other (b) Parallel to the direction of propagation
(c) **Perpendicular to each other** (d) None of these
[2003]
6. Light possesses:
(a) Transverse nature (b) Electromagnetic nature
(c) Dual nature (d) **All of these**
[2001]
7. Which of the following are not electromagnetic waves?
(a) X-rays (b) Radio waves
(c) Ultra-violet rays (d) **α - rays**
[2001]
8. Photoelectric effect proves that light consists of:
(a) **Particles** (b) Waves
[2011, 05]
9. The wave theory of light cannot explain:
(a) Polarization (b) **Photoelectric effect** (c) Interference (d) Dispersion
[2004]
10. The characteristic property of light which does not change with the frequency of light is
(a) Frequency (b) Wavelength (c) **Velocity**
[2013, 03, 02]
11. A monochromatic beam of light is entering from one medium into another. The property which remains unchanged is:
(a) Amplitude (b) Velocity (c) **Frequency** (d) Wavelength

12. The property of light waves that leads to the phenomenon of color is their:
(a) Velocity (b) Amplitude (c) Quality (d) **Wavelength**
13. In the complete electromagnetic spectrum which of the following has least frequency?
(a) X-ray (b) γ rays (c) Ultraviolet light (d) **Infrared light**
14. The wavelength $1000\text{\AA}$ belong to:
(a) Infrared spectrum (b) **Ultra violet spectrum**
(c) Visible range (d) Green light
15. If Planck's constant is $h = 6.625 \times 10^{-34} \text{ J s}$, what energy is associated with light with a wavelength of 160 nm ?
(a) **$1.24 \times 10^{-18} \text{ J}$** (b) $1.88 \times 10^{-15} \text{ J}$ (c) $1.24 \times 10^{-15} \text{ J}$
- [2013]
16. Monochromatic yellow light is unable to show:
(a) Reflection (b) Refraction (c) **Desperation** (d) Interaction
- [2012]
17. Hygen's Principle is used to
(a) Determine the speed of light (b) Expressed polarization
(c) **locate the wave from** (d) find the refractive Index
18. The locus of all points in the same state of vibration is known as:
(a) Half period zone (b) A half wave zone (c) **A wave front** (d) None of the above
- [2007]
19. Wave front near the point source is:
(a) Plane (b) Cylindrical (c) **Spherical** (d) Conical
- [2013, 11, 06, 03]
20. Light year is a unit of:
(a) Energy (b) Time (c) **Distance** (d) Intensity
21. Which is invalid for a photon?
(a) **Its mass is h^2v^2** (b) Its momentum is $h\nu/c$
(c) All its energy is K.E. (d) It has zero rest mass
22. Which of the following phenomenon cannot be explained on the particle nature of light:
(a) Photoelectric effect (b) Compton's effect (c) Pair production (d) **Interference**
23. Two light waves meet at time when one has the instantaneous amplitude A and the other has the instantaneous amplitude B. their combined amplitude is:
(a) $A + B$ (b) **Between $(A + B)$ and $-(A + B)$**
(c) $A - B$ (d) Indeterminate
24. Double slit arrangement is suggested by Young is order to obtain:
(a) Monochromatic light (b) Destructive interference
(c) Constructive interference (d) **Phase coherence**
- [2014, 2004]
25. In Young's double slit experiment, the fringe spacing is:
(a) $\frac{d\lambda}{L}$ (b) $\frac{\lambda L}{d}$ (c) $\frac{d}{\lambda L}$ (d) $L \lambda d$
26. Two monochromatic waves of same wave length are traveling through a medium. They can interfere destructively provided their path difference is:
(a) 2λ (b) λ (c) **$\frac{5}{2}\lambda$** (d) 5λ

27. The phenomena of interference of light was first demonstrated by:
(a) Newton (b) Einstein (c) **Thomas Young** (d) Michelson

28. In Young's double slit arrangement, the bright fringes obtained are of:
(a) **Uniform intensity** (b) Non uniform width (c) Colored (d) Circular

[2001]

29. In Michelson interferometer, semi silvered plate is used to obtain.
(a) Dispersion (b) **Phase coherence**
(c) Monochromatic light (d) Unpolarized light

[2005]

30. To replace a bright fringe by the next bright fringe in a Michelson interferometer, the movable mirror is moved through a distance equal to:
(a) λ (b) $\lambda/2$ (c) $\lambda/4$ (d) 2λ

31. Interferometers measures:
(a) **Wavelength of light** (b) Thickness of thin objects
(c) Illuminating power of light (d) Velocity of light in gases

[2003, 02]

32. Bending of light around the obstacles is called:
(a) Polarization (b) Interference (c) **Diffraction** (d) Refraction

[2006]

33. If 2000 lines /cm are ruled on a grating. Its grating element is:
(a) 5×10^{-4} m (b) 5×10^{-5} m (c) **5×10^{-6} m** (d) 5×10^{-8} m

[2014, 2001]

34. Diffraction of light is special type of.
(a) Reflection (b) Refraction (c) **Interference** (d) Polarization

[2000]

35. When both the point source and the screen are placed at finite distance from the diffracting obstacle the phenomenon is called.
(a) **Fresnel diffraction** (b) Fraunhofer diffraction

36. In Fraunhofer's diffraction wave front used is:
(a) Spherical (b) Circular (c) **Plane** (d) Conical

[2008, 07, 06, 04]

37. The condition for interference in a thin film is reverse because of:
(a) Diffraction (b) **Phase reversal (one part of a wave)**
(c) Refraction (d) Phase coherence

[2005]

38. Colors in thin soap films are due to:
(a) Refraction of light (b) Diffraction of light (c) **Interference of light** (d) Scattering of light

39. In thin films, destructive interference occurs when the path difference is:
(a) Odd multiple of half wavelength (b) Even multiple of wavelength
(c) **Integral multiple of wavelength** (d) None of these

40. Monochromatic green light has a wave length of 520nm in air. The wavelength of this light inside glass of refractive index 1.5 is approximately.
(a) 300nm (b) **346nm** (c) 520nm (d) 780nm

41. When electromagnetic waves strike the boundary of denser medium they are:
(a) Reflected in phase (b) **Reflected out of phase by 180**
(c) Reflected (d) Completely absorbed

[2003]

42. The dispersion of white light after passing through a prism is due to different
(a) Intensities (b) Amplitudes (c) Temperature (d) wavelength
43. Newton's rings are formed due to the phenomenon of
(a) Reflection (b) Thin film interference (c) Diffraction (d) Refraction

[2008]

44. In Newton's rings the central ring is:
(a) Always bright (b)) Can be bright or dark (c) Of blue color (d) Always dark
45. The path difference corresponding to a phase difference of π radian is:
(a) 2λ (b) $\lambda/2$ (c) $\lambda/4$ (d) λ
46. The two light waves can interfere destructively if:
(a) They reach a point in phase (b) They reach a point out of phase by π
(c) They are traveling through a medium in opposite direction (d) None of the above is true
47. X-ray diffraction can be observed by using:
(a) Diffraction grating (b) Rock salt crystal
(c) Convex lens (d) Michelson's interferometer
48. Which of the following equations represents the Bragg's law?
(a) $m\lambda = 2d \sin\theta$ (b) $m\lambda = d \sin\theta/2$
(c) $2m\lambda = d \sin\theta$ (d) $m\lambda = d \sin\theta$

[2007]

49. Diffraction of X-rays can be studied by using:
(a) A thin film (b) Diffraction grating (c) Rock salt (d) None

[2008 ,07, 03, 02, 01]

50. The phenomenon of light which proves that light waves are transverse is:
(a) Reflection (b) Refraction (c) Interference (d) Polarization
- [2005]
51. Which property of light is used to determine the concentration of an optically active substance such as sugar?
(a) Interference (b) Dispersion (c) Diffraction (d) Polarization

[2001]

52. Polarization of light due to tourmaline crystals takes place because of
(a) Reflection (b) Selective absorption (c) Refraction
53. Which of the following is not associated with light waves:
(a) Diffraction (b) Longitudinal vibrations
(c) Interference (d) Transmission of energy

10.1: GEOMETRICAL OPTICS

[2008]

1. In Galilean telescope, the final image formed is:
a) Real and inverted
b) Virtual and inverted
c) Real and erect
d) Virtual and erect
 2. If an astronomical telescope has an objective of focal length 90 cm and the focal length of its eyepiece is 10cm. the length of the telescope will be:
a) 9 cm
b) **100 cm**
c) 80 cm
d) None of these
 3. A concave lens bends the light rays from:
a) **Its optical axis**
b) its optical centre
c) Its Radius
d) none of the above
 4. The defect of lens which can be easily corrected by reducing its aperture is known as:
a) **Spherical aberration**
b) Chromatic aberration
c) Hypermetropia
d) Astigmatism
 5. Chromatic aberration is caused by:
a) Reflection
b) **Dispersion**
c) Refraction
- [2013]
6. Power of a magnifying glass having focal length 5 cm is:
a) 5 Diopter
b) 10 Diopter
c) **20 Diopter**
d) 50 Diopter
- [2008]
7. The power of a concave lens of focal length 50cm is:
a) 0.5 dioptre
b) **- 2 dioptres**
c) 2 dioptres
- [2011]
8. The magnifying power of a glass of focal length 25 cm is:
a) $\frac{1}{2}$
b) 1
c) **4**
d) 2
 9. The unit of the power of a lens is:
a) Watt
b) Joules
c) **Dioptre**
d) Newton
 10. If an object is placed at 2F of a convex lens, the image will be formed at:
a) **2 F**
b) 4 F
c) 3 F
d) F
 11. A student is wearing the glasses of power 2.5 dioptres. This corresponds to the focal length of:
a) 2.5 cm
b) 60 cm
c) 50 cm
d) **40 cm**
 12. Two convex lenses, of the same focal length f, are kept touching each other. The focal length of the combination will be:
a) f
b) **$f/2$**
c) 2 f
d) $2f + 2$
 13. The power of a convex lens of focal length 50 cm is:
a) $\frac{1}{2}$ dioptre
b) **2 dioptre**
c) 1/50 dioptre
d) 50 dioptre
 14. An astronomical telescope when focused for infinity with $f_o = 60$ cm and $f_e = 3$ cm has its length equal to:
a) **63 cm**
b) 20 cm
c) 57 cm
d) 180 cm

15. A convex lens is:
a) Thicker in the middle and thinner on the edge b) Thinner in the middle and thicker
16. The point in the lens through which the light rays will pass without any deviation is called its:
a) Optical centre b) Optical axis c) Principal focus
17. In converging lens when the object is placed within its focal length then the image will form:
a) At its principal focus on the other side **b) On the same side where the object is placed**
18. In convex lens the image distance is taken as positive for
a) Real image b) Virtual image c) None of the above
19. The relation between the focal length and the power of a lens is that
a) They are reciprocal to each other b) They are inversely proportional to each other
20. If the focal length of the lens is 10cm then its power in Diopter is
a) 9 b) 5 c) **10**
21. The focal length of the magnifying-glass is 5cm then its magnification is
a) **6** b) 2 c) 1
22. If the magnification of the lens is 6 and the image distance is 25 cm then the object distance is
a) **4.16 cm** b) 5.16 cm c) 6.16 cm d) 7.16 cm
23. In compound microscope the final image formed is
a) Virtual and diminished b) Real and magnified c) **Virtual and magnified**
24. In compound microscope the objective will form a image which is
a) Virtual and magnified b) Real and diminished c) **Real and magnified**
25. The objective of the astronomical telescope will form an image which is:
a) Virtual and magnified b) Real and magnified c) **Real and diminished**
26. The Galilean telescope consists of
a) Single lens b) **Two lenses** c) Three lenses
27. The magnifying power of the terrestrial telescope is the same as
a) Magnifying glass b) Compound microscope c) **Astronomical telescope**
[2013]
28. In the terrestrial telescope, the central lens is used to:
a) **Erect the image** b) Increase magnifying power c) Both of these d) None of these
29. The length of the astronomical telescope which is focused for infinity is:
a) $f_o - f_e$ b) **$f_o + f_e$** c) None of the above.
30. An astronomical telescope has the magnifying power 100 and focal length of eye piece is 4cm, then the focal length of the objective is
a) 40 cm b) **400 cm** c) 25 cm
[2012, 14]
31. The length of the Galilean telescope when focused for infinity is
a) $f_o - f_e$ b) $f_o + f_e$ c) $f_e - f_o$ d) $\frac{f_o}{f_e}$
32. A telescope has an objective of focal length 60cm and eye piece of focal length 3 cm then its magnifying power is
a) 2 b) **20** c) 180
33. A Galilean telescope has an objective of focal length 25 cm and an eye-piece of focal length 10 cm then its length is given as:
a) 35 cm b) **15 cm** c) 250 cm

34. The instrument which is used to study the spectrum of luminous bodies is called:
a) **Spectrometer** b) compound microscope c) Polar meter
35. Collimator, telescope and turntable are the parts of
a) Compound microscope b) Polar meter c) **Spectrometer**
36. The part of the spectrometer named as Collimator consists of a
a) **Converging lens** b) Diverging lens c) None of these
37. The spectrometer is an analyzing instrument used primarily to discover and measure the
a) Speed of the light b) **wave length of light** c) Frequency of light
38. The spectrum of light which shows band instead of lines is called
a) Continuous spectrum b) line spectrum c) **Band spectrum**
39. The front of the eye is covered by a transparent membrane called:
a) **Cornea** b) Retina c) None of the above
40. The abnormality myopia can be corrected by using.
a) Converging lens b) **Diverging lens** c) None of the above
41. The abnormality hyperopia can be corrected by using a
a) **Converging lens** b) Diverging lens c) None of the above.
42. Under normal condition the human eye is most sensitive for
a) **Red light** b) Yellow – green light c) Blue light
43. The iris controls the:
a) wave-length of light entering in the eye b) Speed of the light entering in the eye
c) **Amount of the light entering in the eye**
44. A lens is a piece of transparent material that can focus:
a) **Transmitted beam of light** b) reflected beam of light c) None of the above
45. The use of a single converging lens is such a position when the object is placed within its focal length is called
a) **Magnifying glass** b) Compound microscope c) None of the above d) Astronomical telescope
46. In order to get higher magnification by magnifying glass, the lens used is of:
a) **Short focal length** b) Long (large) focal length c) None of the above
47. If an object is placed at principle focus 'F' of a converging lens, the image will be formed:
a) at F b) between focus & optical centre c) **at-infinity** d) at 2F